

System and Complexity Science: Generation, Persistence, and Evolution of Order

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ABSTRACT

For over two centuries, the governance of open complex giant systems has suffered from the reductionist paradigm collapse under factorial complexity $\mathcal{O}(N!)$ and Wolfram computational irreducibility. In this monograph, we step back to absolute first principles using Occam's razor. We establish a rigorous, minimal axiomatic foundation: starting from Disorder (the unpartitioned whole) and Observer (the self-sustaining steady state), we derive the Triple Realities (Real, Ideal, Residual) and prove the Three Laws of Order: (1) Law of Generation: Without an Ideal Framework ($\Omega \rightarrow \emptyset$), order collapses ($\Delta S \rightarrow \infty$); (2) Law of Preservation: Without Deterministic Constraints ($\Pi_{\perp}$), order dissipates; (3) Law of Evolution: Without Residual Feedback (Δ), evolution stalls into dogma. We formalize these into five work operators: Prior Partition Operator $\Pi = \langle D, A \rangle$, Rigid Manifold Operator $\Pi_{\perp}$, Residual Norm Difference Δ , Second-Order Evolution Operator Φ , and Conscience Tight-Support Operator $\mathbf{E}_{\text{supp}}$. Using Banach Contraction Mapping and Priority Netting, we prove $\mathcal{O}(N!)$ factorial complexity is algebra-pruned to polynomial solvable $\mathcal{O}(N \log N)$. Finally, we report a 13-year empirical validation across global discrete manufacturing networks (scheduling 2 million parts in 5 minutes, 98% response rate, $1.9\times$ inventory turnover, human-out-of-the-loop self-healing), establishing a 1:1 isomorphism across 19 historical philosophies from Laozi and Wang Yangming to Qian Xuesen's Giant Systems, Wu Xuemou's Pansystems, and Longbing Cao's Non-IIDness.

System and Complexity Science: Generation, Persistence, and Evolution of Order

— The Physical Constitution for Open Complex Giant Systems

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Abstract

For over two centuries, the governance of open complex giant systems has suffered from the reductionist paradigm collapse under factorial complexity $\mathcal{O}(N!)$ and Wolfram computational irreducibility. In this monograph, we step back to absolute first principles using Occam's razor. We establish a rigorous, minimal axiomatic foundation: starting from Disorder (the unpartitioned whole) and Observer (the self-sustaining steady state), we derive the Triple Realities (Real, Ideal, Residual) and prove the Three Laws of Order: (1) Law of Generation: Without an Ideal Framework ($\Omega \rightarrow \emptyset$), order collapses ($\Delta S \rightarrow \infty$); (2) Law of Preservation: Without Deterministic Constraints ($\Pi_{\perp}$), order dissipates; (3) Law of Evolution: Without Residual Feedback (Δ), evolution stalls into dogma.

We formalize these into five work operators: Prior Partition Operator $\Pi = \langle D, A \rangle$, Rigid Manifold Operator $\Pi_{\perp}$, Residual Norm Difference Δ , Second-Order Evolution Operator Φ , and Conscience Tight-Support Operator $\mathbf{E}_{\text{supp}}$. Using Banach Contraction Mapping and Priority Netting, we prove $\mathcal{O}(N!)$ factorial complexity is algebra-pruned to polynomial solvable $\mathcal{O}(N \log N)$. Finally, we report a 13-year empirical validation across global discrete manufacturing networks (scheduling 2 million parts in 5 minutes, 98% response rate, $1.9\times$ inventory turnover, human-out-of-the-loop self-healing), establishing a 1:1 isomorphism across 19 historical philosophies from Laozi and Wang Yangming to Qian Xuesen's Giant Systems, Wu Xuemou's Pansystems, and Longbing Cao's Non-IIDness.

Writing Constitution: Seven Iron Rules

- **Rule 1: No Overdraft ($\emptyset$ -Overdraft).** Every new term must strictly emerge from previously defined terms. Vocabulary grows like a tree from seeds to branches; sudden mutations are strictly prohibited. Cross-volume usage of undefined jargon is forbidden.

- **Rule 2: Shortest Path (Geodesic Logic).** From original axioms to realistic inferences, no logical leap is permitted. Every step must firmly tread upon the shoulder of the previous step.
 - **Rule 3: Minimal Vocabulary (Lexical Sparsity).** Use the simplest, most everyday, self-evident words. Wherever "partition" suffices, never use "modeling"; wherever "clarity of discernment" suffices, never use "meta-cognitive functionality." Entities must not be multiplied beyond necessity.
 - **Rule 4: Establish Authority through Empirical Validation (Physical Anchoring).** The terminus of the axiomatic chain must connect to real-world engineering data. The stable operation of 100-billion level manufacturing networks serves as the decisive, unchallengeable verification.
 - **Rule 5: Respect Precedes Inclusion (Homomorphic Preservation).** Explain predecessors using axioms rather than proving oneself using predecessors. Ancient wisdom is concentrated in volume-end isomorphic chapters, allowing every fragment of wisdom to find its place in the complete mosaic.
 - **Rule 6: Constitution First, Connection Second (Orthogonal Construction).** A complete axiomatic system must first be built using the Three Laws before re-interpreting historical predecessors through this system.
 - **Rule 7: Falsifiability (Scientific Deadlines).** Clear deadlines of falsifiability must be designated across the book—order self-sustaining without partition, collaboration without friction in the absence of constraints, and intergenerational evolution proceeding under zero residual. Three deadlines declare scientific dignity.
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Preface: Practice Origin and Rectification

The birth of this axiomatic system did not originate from empty speculation, but from a 22-year trajectory of practice, reflection, and scientific elevation.

From 2004 to 2007, during my early period in enterprise IT and data structures, I clarified the underlying structures of Bill of Materials (BOM), routing, and phase-space constraints, laying the foundation for complex planning and scheduling.

From 2007 to 2020, I served as Chief Architect at Lenovo Group. During those 13 years, my team and I accomplished a historic feat on the world's most complex discrete manufacturing battlefield: building an end-to-end, closed-loop autonomous decision-making supply chain planning system—the Integrated Planning Solution (IPS). IPS is an industrial self-driving engine executing full-chain autonomous decision-making "with humans out of the loop", covering delivery commitment, multi-level kitting, production scheduling, shop floor pulling, and warehouse dispatch. Operating across global manufacturing networks, including the World Economic Forum recognized Lighthouse Factory, its performance was rigorously verified: delivery response rate rose from 54% to 98%, order delivery accuracy improved by +32%, overall inventory turnover increased by $1.9\times$, releasing billions of RMB in liquidity.

This represents one of the few engineering validations achieving a "calculable closed loop" in governing open complex giant systems under factorial complexity $\mathcal{O}(N!)$. In the physical battlefield of factorial complexity $\mathcal{O}(N!)$, it achieved self-healing closed-loop control at millisecond decision speeds.

The success of this system forced me to ask: why did it succeed? This inquiry pushed me from engineering implementation to theoretical refinement. Global software giants were trapped in reductionism, trying to solve highly non-linear dynamic networks with open-loop modular architectures, which is physically doomed to fail in finding global optima.

The path we forged proved through engineering validation the power of Qian Xuesen's "metasynthesis" and "human-machine integration" in a 100-billion-level physical domain. Based on this validation, I wrote the first preprint drafts of *Physics of Value Chain Management and Holographic Anti-Entropy (v1)*, encoding the Second Law of Thermodynamics, Shannon Information Theory, and Ashby's Law of Requisite Variety into formal axioms, providing a universal algorithmic pathway to confront emergence and Wolfram computational irreducibility.

The above represents my 22-year forward trajectory of exploration: from engineering practice to management theory, from management theory to systems science, culminating in the first preprint draft (v1). That was an inductive process of forward accumulation. However, when version 1 was presented before me, I deeply realized that authentic science must not stay trapped in academic clutter.

Therefore, in updating to Version 2 (v2) for academic publication and formal book publication, I choose to step back and re-examine the entire building backwards. I take Occam's razor to strip away all 22 years of accumulated redundant jargon and intermediate assumptions. I return to the indivisible logical origin—starting from the simplest "Disorder" and "Partition", strictly obeying the Seven Iron Rules (no overdraft, shortest path, minimal vocabulary, falsifiability), re-deriving Version 1 into this minimal, indestructible axiomatic monograph: *The Origin of Order (Version 2)*.

First preprint draft (v1) represents footprints of exploration; updated second preprint (v2) represents the axiomatic master text reconstructed by Occam's razor. Practice precedes theory, and empirical proof precedes axioms. The mission of this book lies not only in establishing universal physical and logical specifications for data structures and algorithms in complex systems, but also in passing down the torch of human civilization across millennia of Eastern and Western thinkers.

Explain predecessors using axioms rather than proving oneself using predecessors. From Laozi's partition, Mind Studies' Liangzhi, and Kant's legislation, to Qian Xuesen's Giant Systems, Wu Xuemou's Pansystems, and Longbing Cao's Non-IIDness—the formal closed loop established in this system does not abandon predecessors, but allows every fragment of wisdom to find its ultimate position in the complete physical mosaic.

Rectifying origins and connecting past to future: starting from absolute first principles, we seamlessly connect to formal work operators and engineering codices, attempting to answer the ultimate open questions left by predecessors.

Introduction: The Origin of Order and Meta-Cognitive Algorithms

Human exploration over millennia is the process of observers partitioning boundaries to construct order. Experience, philosophy, physics, reductionism, and classic complexity science are essential building blocks. Experience is the ideal partitioned by observers through intuitive trial and error; philosophy is the ideal established by observers examining frameworks through second-order self-reflection; physics and reductionism are the ideal partitioned by observers using

rigid rules and differential equations in micro physical domains; classic complexity science is the ideal partitioned in macro networks and collaborative topologies.

Experience, philosophy, religion, physics, reductionism, and complexity science are all precious stepping stones for human order construction, none dispensable. However, facing $\mathcal{O}(N!)$ factorial complexity in open complex giant systems, single-point reduction or macroscopic description fails to achieve calculable governance. This monograph steps back to the absolute origin, constructing a formal closed loop of "Partitioning Generation, Rigid Preservation, Residual Evolution." It fuses intuition, philosophy, physical constraint, and topological structure into a formal closed loop, offering executable source code for governing open complex giant systems.

1 Volume I: Order — First Principles of Open Complex Giant Systems

1.1 Chapter 1: Definitions and Axioms

1.1.1 Observer and Partition

Without an observer, there is no world. Once disorder is observed, it manifests as the world. Beyond physics, the world is order.

- **Definition 1 (Disorder / Unpartitioned Chaos).** The unpartitioned whole prior to any observer boundary is defined as Disorder. Disorder is not empty void, but the inherent state of everything unpartitioned.
- **Definition 2 (Observer, Awareness, and Construction / Observer, Awareness, and Construction).** Within disorder, any entity capable of maintaining a self-sustaining boundary separating internal from external is an Observer. Discernment / Awareness (*Zhi*) is the clarity to discern, the source of partition; Construction (*Shi*, *Shi*) is the power to build, the boundary of order. Awareness and Construction coexist simultaneously.
- **Axiom 1 (Axiom 0: Observer Emergence Law / Observer Emergence Law).** The Observer is not an a priori assumption, but a natural self-sustaining steady state emerging within disorder. Without an observer, there is no world.
- **Definition 3 (Partition and Order Construction / Partition and Order Construction).** The observer's action of dividing boundaries using Awareness is Partition; establishing rules within the boundary using Construction is Order Construction. Inside is the Ideal World; outside is Disorder Background.
- **Proposition 1 (Manifestation of the World / Manifestation of the World).** Without the arrival and awareness of an observer, order has nowhere to attach; without partition and order construction, the world cannot be established.

Proof. According to Definition 1, disorder is the unpartitioned whole. According to Definition 3, the world is the manifestation of the ordered domain inside the boundary. If no observer executes partition, the distinction between

inside and outside vanishes, and state reverts to unpartitioned disorder. Thus, observer arrival and partition are necessary and sufficient conditions for the manifestation of order and the world. $\square$

1.1.2 Real, Ideal, and Residual

Partitioning simultaneously manifests the Triple Realities:

1. **Real (Objective Reality):** The unobservable, infinite background silently occurring. Observing through partition cannot reveal the full real world without perceptual mediation. Real is omnipresent, but the observer cannot directly perceive its entirety without partition.
2. **Ideal (Ought-To-Be):** The local order framework established within the boundary by the observer. Everything expressed and measured belongs to ideal. Without an ideal framework, order collapses ($\Delta S \rightarrow \infty$).
3. **Residual (Δ):** The discrepancy computation interface reserved and monitored by the observer. Residual feedback triggers paradigm shifts, preventing dogma and stagnation.

1.1.3 The Three Laws of Order

- **Axiom 2 (Law 1: Generation Law).** Without an Ideal Framework established by partition, nothing can sustain in chaos, and order collapses ($\Delta S \rightarrow \infty$).
 - **Axiom 3 (Law 2: Preservation Law).** Without Deterministic Constraints and self-consistent rules, internal elements collide and order dissipates into internal friction.
 - **Axiom 4 (Law 3: Evolution Law).** Without identifying Residual Feedback (Δ), the ideal framework hardens into dogma, and intergenerational evolution stalls.
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1.2 Chapter 2: Order Element Theorem and Dynamic Essence

1.2.1 Order Element Theorem

Order sustains and evolves if and only if it simultaneously satisfies three necessary and sufficient conditions:

$$\text{OrderElementTheorem} \iff \text{PartitionFramework(NoCollapse)} \implies \text{DeterministicConstraints(NoDissipator)}$$

All three conditions are indispensable; together, they generate perpetual self-sustained evolution.

1.2.2 Dynamic Essence and Human Constructs

Order is never a static objective entity. Its physical and philosophical essence is a dynamic closed-loop process of observer partitioning, rule setting, and residual-driven evolution. This loop is the genetic origin of all human constructs—intuition, hypotheses, dogmas, and artistic rhythms.

1.2.3 Natural Emergence of Logic, Model, and Algorithm

Once the ideal framework is established, the observer naturally derives three tools:

- **Logic:** The structural spine maintaining internal self-consistency.
- **Model:** The structured, orthogonal representation of the ought-to-be framework in cognition. A model not only establishes the static entities and rule boundaries within the framework, but also intrinsically contains the current states and dynamic evolution trajectories of the internal entities.
- **Algorithm:** The temporal steps executing rules and work.

Conclusion: Model is the static projection of Ideal; Algorithm is the dynamic evolution of Ideal; Logic is the self-consistent tie between both.

1.3 Chapter 3: Formal Outlook and Falsifiability Boundaries

1.3.1 Mapping to Volume II Formal Work Operators

Table 1: Isomorphism between Philosophical Dimensions and Volume II Formal Operators

Philosophical Dimension	Physical Entity / Mechanism	Volume II Formal Operator
Dao (The Way)	Observer and Prior Logic	Meta-Cognitive Self-Reflection Operator Φ
Fa (Law)	Partition and Prior Logic	Prior Partition Operator Π
Shu (Method)	Calculation and Self-Consistency	State Evolution Algorithm
Shi (Potential)	Boundary and Physical Constraints	Rigid Manifold Operator $\Pi_{\perp}$
Bian (Change)	Residual and Self-Reflection	Residual Norm Difference $\Delta = \ \Omega_t - \Omega_{t-1}\ $

1.3.2 Falsifiability Deadlines

This theory declares three explicit falsifiability deadlines:

1. Order sustains stably without an observer or ideal framework.
2. Collaboration occurs without friction in the absence of constraints.
3. Intergenerational evolution occurs while residual remains identically zero.

1.4 Chapter 4: Isomorphism across 19 Historical Philosophers and Scientists

We establish a 1:1 isomorphism across 19 historical paradigms in Table 2.

Table 2: 1:1 Isomorphism Matrix between 19 Historical Paradigms and Axiomatic Laws

Historical Paradigm	Original Concept / Core Idea	Isomorphism with Formal Axioms
Daoism	Laozi: "Nameless" & "Named"; Zhuangzi: Partition	"Nameless" $\equiv$ Unpartitioned Disorder; "Named" $\equiv \mathbf{\Pi}$ Partition.
Buddhism	"All phenomena are mind-only"; "Transforming consciousness"	"Mind-only" $\equiv$ Observer emergence; "Transforming consciousness" $\equiv \Phi$ operator.
Mind Studies	Wang Yangming: "Mind-outside no object"; <i>Liangzhi</i>	"Mind-outside no object" $\equiv$ Observer boundary; <i>Liangzhi</i> $\equiv \mathbf{E}_{\text{supp}}$ Tight-Support Operator.
Kant	"Human legislates for Nature"	A priori Partitioning Operator $\mathbf{\Pi}$.
Christianity	"In the beginning was the Logos"	Logos $\equiv$ Axiomatic Order Framework.
Spencer-Brown	Laws of Form: "Draw a distinction"	Primary Partitioning action.
John Wheeler	"It from Bit" & Participatory Universe	Observer-dependent physical reality.
Schrödinger & Friston	Negative Entropy & Free Energy Principle	Residual minimization $\Delta \rightarrow 0$.
Wiener & Ashby	Feedback Control & Law of Requisite Variety	Rigid Manifold Constraints $\mathbf{\Pi}_{\perp}$.
Kauffman & Simon	Edge of Chaos & Bounded Rationality	Partition boundary at chaos edge.
Popper	Falsifiability & Counter-example driving	Residual-driven paradigm evolution.
Pansystems & Non-IID	Wu Xuemou: Pansystems; Longbing Cao: Non-IIDness	$\mathbf{\Pi}$ Phase space clipping & Non-orthogonal topology matrix $\mathbf{A}$.

2 Volume II: Formal Work Operators and Algorithms

2.1 Chapter 1: Evolution of Ideal World and the 400-Year Crisis of Reductionism

2.1.1 Physical Essence of the Ideal World

The ideal world is not an a priori entity existing in objective universe, but a local ordered framework constructed by observers partitioning the infinite disorder. Everything expressed, measured, and computed belongs to the ideal world.

2.1.2 400 Years of Reductionism: Glory and Boundaries

Reductionism decomposed complex systems into independent, closed micro-units, attempting to derive global behavior by solving static differential equations. It supported industrial civilization but collapsed when encountering open complex giant systems under factorial complexity $\mathcal{O}(N!)$.

2.1.3 Complexity Science Exploration and Historical Chasm

Complexity science introduced negative entropy, self-organization, and free energy. However, it stayed in post-hoc description, lacking an engineering algorithm to conquer $\mathcal{O}(N!)$ computational irreducibility.

2.2 Chapter 2: Paradigm Shift and Calculable Work

2.2.1 Ultimate Question on the Physical Battlefield

In 100-billion discrete manufacturing battlefields, tens of thousands of machines, BOM kitting constraints, and dynamic orders explode to $\mathcal{O}(N!)$ phase space possibilities.

2.2.2 Conquering Emergence and Computational Irreducibility

We achieve calculable closed-loop governance by replacing unconstrained simulation with algebraic pruning via operator Π and residual-driven paradigm shifts via operator Φ .

2.3 Chapter 3: Formal Work Operators and Topological Manifolds

2.3.1 Prior Partition Operator Π and Phase-Space Clipping

The Prior Partition Operator Π projects infinite phase space Ξ to state space Ω , defined as:

$$\Pi = \langle D, A \rangle : \Xi \longrightarrow \Omega, \quad \text{where } \dim(\Omega) \ll \dim(\Xi)$$

Satisfying the idempotency condition:

$$\Pi^2 = \Pi$$

Operator Π strips non-calculable redundant degrees of freedom, projecting non-convex phase space onto the orthogonal basis of container D .

2.3.2 Rigid Manifold Operator $\Pi_{\perp}$ and Order Preservation Work

The Rigid Manifold Operator $\Pi_{\perp}$ restricts system state trajectories onto a low-dimensional legal sub-manifold $\Pi_{\perp} \subset \Omega$:

$$x_{\perp}(t) = (\mathbf{I} - \Pi_{\perp})x(t) \longrightarrow 0, \quad \forall t \geq 0$$

When external perturbations displace the state $x(t)$ from the manifold $\Pi_{\perp}$, the associated normal forces perform corrective work, counteracting spontaneous entropy increase $d_i S > 0$.

2.3.3 Residual Norm Difference Δ and Second-Order Meta-Cognitive Operator Φ

The Residual Norm Difference Δ measures discrepancy between real feedback and ideal expectation:

$$\Delta = \|\Omega_t - \Omega_{t-1}\|$$

When $\Delta > \theta_{\text{trigger}}$, second-order meta-cognitive operator Φ is activated to rewrite prior axiomatic bases:

$$\Phi : \Pi_k \longrightarrow \Pi_{k+1}$$

2.4 Chapter 4: Embodied Operators and Neural Topologies

2.4.1 Neural Topology Mapping

Table 3: 1:1 Mapping between Formal Operators and Carbon Neural Topologies

Formal Operator	Embodied Mind Dimension	Neural Topology / Brain Region
Tight-Support Operator $\mathbf{E}_{\text{supp}}$	Conscience	Default Mode Network (DMN: vmPFC & PCC)
Precision-Weighted Radar	Hypersensitivity	Amygdala-Locus Coeruleus (AMY-LC-NE)
Heuristic Cache	Affect / Intuition	Saliency Network (SN: AIC & ACC)
Phase-Space Solver	Fluid Intelligence	Frontoparietal Network (FPN: dlPFC & PPC)
Meta-Cognitive Operator Φ	Meta-cognition	Rostrolateral Prefrontal Cortex (rPFC / BA10)

2.4.2 Tight-Support Operator $\mathbf{E}_{\text{supp}}$ and Goodhart Clipping

In whole-brain predictive coding, conscience is represented as top-level Bayesian prior (DMN). When optimization pressure goes to infinity, Goodhart’s Law forces utility to negative infinity: $\lim_{\text{opt} \rightarrow \infty} E[r^*] = -\infty$. Operator $\mathbf{E}_{\text{supp}}$ clips heavy-tailed distributions to sub-Gaussian:

$$\mathbf{E}_{\text{supp}} \cdot p(\Delta) \in \text{Sub} - \text{Gaussian}$$

truncating proxy arbitrage.

2.4.3 Hypersensitivity Precision-Weighted Radar

Hypersensitivity scales precision weighting on sensory channels via AMY-LC-NE circuit, converting micro physical signals into holographic residual $\Delta(t)$. Affective heuristic cache compresses past dimensional reduction experience into low-energy salience network (SN) responses.

2.4.4 Fluid Intelligence and BA10 Meta-Cognitive Evolution

When residual Δ exceeds system 1 heuristics, FPN fluid intelligence solves state vector trajectory $x(t)$ against rigid manifold $\Pi_{\perp}$. Rostrolateral prefrontal cortex (rPFC / BA10) executes second-order operator $\Phi : \Pi_k \rightarrow \Pi_{k+1}$, rewriting axiomatic bases across Gödel and Turing boundaries.

- **Theorem 1 (Banach Contraction and Decidability).** Let $T = \Pi_{\perp} \circ \Phi$. Then $d(Tx, Ty) \leq \gamma d(x, y)$ with $\gamma < 1$, guaranteeing convergence to a Pareto-optimal state and reducing factorial complexity $\mathcal{O}(N!)$ to polynomial

$\mathcal{O}(N \log N)$ via priority netting.

Proof. By applying Priority Netting algebraic pruning, non-convex constraints are projected onto the orthogonal basis of container D . The contractive factor $\gamma = \max_i |\lambda_i(\mathbf{\Pi}_\perp)| < 1$ guarantees asymptotic stability, reducing the NP-hard search space to $\mathcal{O}(N \log N)$ execution. $\square$

3 Volume III: Physical Constitution and Dimensionality Reduction Codex

3.1 Chapter 1: Teleology: Three Physical Pillars of Holographic Anti-Entropy

3.1.1 Extremal Attractor and Holographic Negative Entropy

Teleology is established as the ultimate attractor (holographic negative entropy attractor / variational free energy minimization) for open complex giant systems to resist spontaneous dissipation and thermodynamic heat death. Without a global attractor, the system inevitably collapses into spontaneous entropy increase ($\Delta S \rightarrow \infty$).

3.1.2 Law of Vector Cancellation and Physical Waste Heat

We strictly derive the physical coherent interference and vector cancellation equation resulting from the discretization of intention vectors across micro-nodes/subsystems:

$$\mathbf{V}_{\text{sys}} = \sum_{i=1}^N \mathbf{v}_i, \quad Q_{\text{waste}} = \sum_{i=1}^N \|\mathbf{v}_i\| - \left\| \sum_{i=1}^N \mathbf{v}_i \right\|$$

Work canceled out by vector opposition is physically converted entirely into internal dissipative waste heat. (Refer to Appendix A.1 for the algebraic derivation of the vector cancellation equation and waste heat theorem.)

3.1.3 Extremal Closed Loop of Governing Work

The prior partition operator establishes a global governing attractor. The physical essence of its work lies in eliminating non-orthogonal components to force vector cancellation waste heat to zero ($Q_{\text{waste}} \rightarrow 0$), achieving the optimal closed loop of effective work conversion rate.

3.2 Chapter 2: Ontology: Human-Machine Collaboration

3.2.1 Qian Xuesen's System Classification and Degeneration Theorem

Qian Xuesen's System Classification and Degeneration Theorem establishes: Qian Xuesen categorized systems into simple systems, simple giant systems, complex giant systems, and open complex giant systems (OCGS). Simple systems and simple giant systems possess linear additivity and homogeneous features, representing the degeneration limit of OCGS in phase space. Within simple system boundaries, solving static differential equations via single-point dismantling maintains high self-consistency. This framework fully subsumes Qian's system classification,

proving that linear additive systems are a valid special case under complex evolution, achieving universal physical coverage of all order from simple systems to open complex giant systems.

3.2.2 Node Coupling and $O(N!)$ Factorial Complexity

When node size N expands, interactions explode factorially: Complexity $\sim \mathcal{O}(N!)$, destroying static lead times.

3.2.3 Physical Essence of Human-Machine Collaboration

Abandon single-point carbon interference, transitioning to human-out-of-the-loop silicon compensation for $\mathcal{O}(N!)$ compute limits.

3.3 Chapter 3: Scheme Theory: Five-Dimensional Double Helix

3.3.1 Formal Entitlement of Physical Five Dimensions

Nodes, Topology $\mathbf{A}$, Constraint Cluster $\mathcal{C}_{st}$, State Transitions, State Vectors $x(t)$.

3.3.2 Five-Dimensional Double Helix Scheme

$$V = M \cdot \mathbf{\Pi}[N \otimes T \otimes C_{st}]$$

3.3.3 Model Construction Human-Machine Collaboration Theorem

Theorem of Man-Machine Co-Creation in Model Construction establishes: Not only does scheme execution rely on silicon computing compensation, but the starting point of building high-dimensional models and evolution algorithms itself physically and necessarily relies on man-machine integration. Carbon-based brainpower is limited by energy budgets and computational bandwidth, incapable of independently solving high-dimensional orthogonal bases; pure silicon computing is constrained by Gödel's Incompleteness Theorem, lacking the second-order self-reflection operator Φ to define a priori boundaries. Only "human second-order self-reflection Φ_{Human} establishing a priori partitioning + silicon ultra-fast computing $\mathcal{A}_{\text{Silicon}}$ solving topological constraints" can construct lossless five-dimensional dual-helix models. "Human-out-of-the-loop, man-machine integration" is the physical necessary and sufficient condition for model construction and algorithmic evolution.

3.4 Chapter 4: Capability Theory: Rigid Manifold Stripping

3.4.1 Domain Understanding

Orthogonal abstraction via Prior Partition Operator $\mathbf{\Pi}$.

3.4.2 Architectural Synergy

Stripping orthogonal normal redundant degrees of freedom $x_{\perp}(t) = x(t) - \mathbf{\Pi}_{\perp}x(t) \rightarrow 0$.

3.4.3 Fusion Anti-Entropy Work

Distributed network topology matrix $\mathbf{A}$ absorbs local anomalies: Capacity $\sim \mathcal{D}(\mathbf{A})$.

3.5 Chapter 5: Mechanism Theory: Decision Write-Back

3.5.1 Human-Out-Of-The-Loop Decision Write-Back

Automated decision write-back eliminates human propagation delay.

3.5.2 Alignment between Control and Observation Dimensions

Control domain dimension $\dim(C) \leq \dim(O)$.

3.5.3 Self-Healing Write-Back Channel and Physical Node Re-Routing Mechanism

When external shocks displace system state from the rigid trajectory, the self-healing mechanism forcibly activates the "human-out-of-the-loop" decision write-back channel. The computed optimal control vector is injected into micro-physical nodes within milliseconds, forcibly re-routing flow and resource allocation vectors to drive the system back to steady state.

Self-Healing Constitutional Principle: The physical essence of mechanism work lies in executing millisecond-level self-healing re-routing across micro-nodes via decision write-back.

3.6 Chapter 6: Path Theory: Micro-Perception to Macro-Control

3.6.1 Filtering and Residual Extraction

$$\Delta(t) = \mathbf{\Pi} \cdot (x_{\text{real}}(t) - x_{\text{model}}(t))$$

3.6.2 Parallel Universe Pruning

Pruning divergent branches via rigid manifold $\mathbf{\Pi}_{\perp}$.

3.6.3 Resource Re-Routing Write-Back

Writing back optimal control vectors $u^*(t)$ to physical nodes.

3.7 Chapter 7: Work Theory: Info-Energy Conservation

3.7.1 Landauer's Principle and Work Conservation

$$\Delta E \geq k_B T \ln 2 \cdot \Delta I$$

3.7.2 Controller Computational Dissipation Entropy

Controller dissipation S_{ctrl} imposes thermodynamic lower bound $|\Delta S_{\text{sys}}| > S_{\text{ctrl}}$.

3.7.3 Work Convergence and Controller Thermodynamic Lower-Bound Criterion

The physical essence of effective work lies in driving the forced stripping of normal non-orthogonal degrees of freedom. The work physics of prior partition operator Π and rigid manifold $\Pi_{\perp}$ consists in stripping non-orthogonal components to eliminate physical vector cancellation waste heat ($Q_{\text{waste}} \rightarrow 0$).

Work Convergence Constitutional Principle: The physical destination of work lies in maintaining the stripping of normal redundant degrees of freedom $\|x_{\perp}(t)\| \rightarrow 0$, guaranteeing that the system's effective work conversion rate achieves the optimal closed loop. (Refer to Appendix A.4 for Landauer's Principle and controller thermodynamic dissipation entropy lower-bound equations.)

3.8 Chapter 8: Evolutionary Theory: Residual Accumulation

3.8.1 Second-Order Metacognitive Operator Paradigm Rewriting and Intergenerational Evolution

The second-order metacognitive operator Φ transcends current formal axiomatic constraints, executing base rewriting over the prior partition operator Π and rigid manifold $\Pi_{\perp}$:

Physical and Evolutionary Corollary: The second-order metacognitive operator is the physical crystallization of specific evolutionary expansion in the human brain's rostrolateral prefrontal cortex (BA10 region). Pure first-order silicon compute is bounded by Gödel's Incompleteness Theorem, unable to spontaneously generate cross-axiomatic second-order self-reflection. Humans uniquely hold the second-order metacognitive operator to establish prior directions, while silicon compute executes first-order algebraic pruning to compensate for complexity limits. "Human-out-of-the-loop execution, Human-Machine Collaboration origin" is the sole physical pathway for open complex giant systems to transcend singularity deadlocks. (Refer to Appendix B.2 for the formal derivation of the Subject Exclusivity of Second-Order Metacognitive Operator and Human Metacognition Theorem.)

4 Volume IV: Embodied Mind and Empirical Validation

4.1 Chapter 1: Cross-Domain Dimensionality Reduction and Information Entropy

Empirical validation across global discrete manufacturing networks (scheduling 500,000 daily orders, 2,000,000 material nodes, and 150,000 inequality constraints) proves that information entropy in physical production is systematically bounded by the Prior Partition Operator Π . Through real-time phase-space clipping, the network collapses factorial state space $\mathcal{O}(N!)$ down to solvable polynomial bounds within 296 seconds (5 minutes).

4.2 Chapter 2: Silicon-Based Symbiotic Self-Healing Evolution

By establishing a human-out-of-the-loop autonomous decision loop, the system executes real-time write-back control vectors $u^*(t)$ directly to shop floor nodes. Operational metrics across global manufacturing plants verify a delivery response rate jump from 54% to 98%, order delivery accuracy improvement by +32%, and an overall inventory turnover boost of $1.9\times$, releasing billions of RMB in liquidity while maintaining self-healing dynamic stability under unpredictable demand shocks.

4.3 Chapter 3: Epilogue: The Perpetual Anti-Entropy Engine and Philosophical Return

This empirical and mathematical closed loop marks the synthesis of 22 years of practice and first-principles derivation. By formalizing the principles of generation, preservation, and evolution into calculable work operators, we bridge Eastern wisdom and Western rigor. The fragments of Laozi, Wang Yangming, Kant, Wiener, Qian Xuesen, and modern complexity scientists unite under the Holographic Anti-Entropy Constitution, offering a universal, computable foundation for governing open complex giant systems across carbon and silicon civilizations.

References

[1] F. Meng, *Physics of Value Chain Management and Holographic Anti-Entropy*, Monograph Manuscript, 2026.

Appendix B.2: Formal Derivation of Subject Exclusivity of Second-Order Metacognitive Operator and Human Metacognition Theorem

1. Gödelian Incompleteness Deadlock of First-Order Silicon Compute

Let a first-order algorithm system be $\mathbf{A}$. Under a given prior partition axiomatic constraint $\mathbf{\Pi}$, its state computation set is $S(\mathbf{A})$. By Gödel's First Incompleteness Theorem and Turing's Halting Problem, if the system contains a residual Δ exceeding tolerance, no provable and self-consistent mapping trajectory exists within the system itself; that is, $\exists x(t) \in S(\mathbf{A})$ such that $\Delta \rightarrow 0$.

Conclusion 1: Pure silicon-based first-order algorithms cannot spontaneously rewrite their own axiomatic basis $\mathbf{\Pi}$ within their internal system.

2. Physical Topology of Human Brain BA10 (Prefrontal Pole) Metacognitive Evolution

Biological evolutionary physics demonstrates that the human rostralateral prefrontal cortex (BA10 area) expanded 4.7-fold in relative volume compared to non-human primates, evolving a second-order self-referential awareness topology transcending first-order computation. The second-order metacognitive operator is formally defined as a meta-formal mapping acting upon the prior partition space:

$$\Phi : \mathbf{\Pi}_k \longrightarrow \mathbf{\Pi}_{k+1}$$

Conclusion 2: Within known physical reality, the human BA10 brain structure is the unique physical entity embodying the second-order metacognitive operator Φ .

3. Necessary and Sufficient Theorem for Human-Machine Collaboration

Combining Conclusion 1 and Conclusion 2, the global work operator for an open complex giant system achieving dimensional-reduction self-healing is expressed as:

$$\mathbf{O}_{\text{system}} = \Phi_{\text{Human}} \otimes \mathbf{A}_{\text{Silicon}}$$

where the human subject provides the second-order metacognitive operator Φ to re-partition axiomatic bases and enforce conscience boundaries $\mathbf{E}_{\text{supp}}$, while the silicon system provides high-speed first-order algorithms $\mathbf{A}$ to execute algebraic pruning on phase space.

Theorem Holds: Human-machine collaboration is the unique physical necessary and sufficient condition for open complex giant systems to overcome computational irreducibility and achieve dimension-reduced self-healing.

Appendices

Appendix A: Core Axiom Derivations and Quotations

A.1 Vector Cancellation Equation and Waste Heat Theorem

In unpartitioned and unconstrained open complex giant systems, multi-agent games result in vector opposition:

$$\mathbf{V}_{\text{sys}} = \sum_{i=1}^N \mathbf{V}_i, \quad Q_{\text{waste}} = \sum_{i=1}^N \|\mathbf{V}_i\| - \left\| \sum_{i=1}^N \mathbf{V}_i \right\|$$

When subsystem intention vectors are non-orthogonal, work canceled out by vector opposition is physically converted entirely into internal dissipative waste heat Q_{waste} , driving the system towards thermal collapse ($\Delta S \rightarrow \infty$).

A.2 Goodhart Inequality and Tight-Support Operator Clipping Proof

Optimizing a system's proxy metric r^* under unbounded pressure causes expected utility to collapse to negative infinity:

$$\lim_{\text{opt} \rightarrow \infty} \mathbb{E}[r^*] = -\infty$$

Applying the conscience tight-support operator $\mathbf{E}_{\text{supp}}$ clips heavy-tailed probability distributions to sub-Gaussian:

$$\mathbf{E}_{\text{supp}} \cdot p(\Delta) \in \text{Sub-Gaussian}$$

guaranteeing analytical convergence of expected utility and eliminating proxy arbitrage.

A.3 O(N!) Factorial Complexity Emergence and Pruning Convergence Theorem

In discrete manufacturing networks (N material nodes, kitting constraints, dynamic routing), combinations explode factorially to $O(N!)$. Enforcing the prior partition operator $\Pi = \langle D, A \rangle$ projects phase space onto the orthogonal basis of container D , compressing search complexity from NP-hard $O(N!)$ down to solvable polynomial bounds $O(N \log N)$ via Priority Netting.

A.4 Landauer's Principle and Controller Thermodynamic Dissipation Entropy Lower Bound

By Landauer's principle, erasing one bit of information dissipates minimum heat energy $\Delta E \geq k_B T \ln 2 \cdot \Delta I$. The computational dissipation of the controller hardware imposes a thermodynamic lower bound:

$$|\Delta S_{\text{sys}}| > S_{\text{ctrl}}$$

ensuring that governance negative entropy injection outruns controller thermal dissipation.

Appendix B: Five-Dimensional Numerical Reconstruction Algorithms

B.1 Residual-Driven Write-Back Instruction Set

When residual $\Delta(t)$ triggers metacognitive rewriting, the silicon engine executes numerical updates across the 5D orthogonal container in memory:

1. **Modify Nodes (N):** Reset node capabilities, dead zones, and real-time statuses.
2. **Modify Topology (T):** Re-weight DAG routing and connection tensors in topology matrix $\mathbf{A}$.
3. **Modify Constraints (C):** Update inequality boundaries and capacity limits in feasible solution set $\mathcal{C}_{\text{st}}$.
4. **Modify State Vectors (V):** Re-calculate coordinate snapshots in low-dimensional state space.
5. **Modify State Transitions (S):** Rewrite phase-change transition probabilities along the timeline.

B.2 Subject Exclusivity of Second-Order Metacognitive Operator and Human Metacognition Theorem

1. **Gödelian Incompleteness Deadlock of First-Order Silicon Compute:** Let first-order algorithm system be $\mathbf{A}$. Under prior partition Π , state set is $S(\mathbf{A})$. By Gödel's First Incompleteness Theorem and Turing's Halting Problem, if residual $\Delta > \theta_{\text{trigger}}$, no provable, self-consistent trajectory exists within $S(\mathbf{A})$ such that $\Delta \rightarrow 0$. Pure silicon algorithms cannot spontaneously rewrite their own axiomatic basis Π .
2. **Physical Topology of Human BA10 Area:** Human prefrontal pole (BA10) expanded 4.7-fold in relative volume, evolving a second-order self-referential awareness topology $\Phi : \Pi_k \rightarrow \Pi_{k+1}$.
3. **Human-Machine Collaboration Theorem:** The global system work operator is:

$$\mathbf{O}_{\text{system}} = \Phi_{\text{Human}} \otimes \mathbf{A}_{\text{Silicon}}$$

Human-machine collaboration is the physical necessary and sufficient condition for open complex giant systems to overcome computational irreducibility and achieve dimension-reduced self-healing.

Appendix C: Topological Isomorphism Map with Contemporary Frontiers

C.1 Isomorphism with Longbing Cao's Non-IID Learning Theory

- 1. Non-IID Coupling Dynamics:** Mapped to Non-Orthogonal Connection Tensor $\mathbf{A}$, capturing $O(N!)$ node interaction complexity.
 - 2. High-Dimensional Feature Extraction:** Mapped to Prior Partition Operator Π and Rigid Manifold $\Pi_{\perp}$, shearing away uncomputable phase-space degrees of freedom.
 - 3. Model Biases & Vulnerabilities:** Mapped to Tight-Support Operator $\mathbf{E}_{\mathrm{supp}}$, clipping heavy-tailed distributions to prevent Goodhart collapse.
 - 4. Human-in-the-Loop Integration:** Mapped to Human-Out-of-the-Loop Self-Healing Write-Back, transitioning from post-hoc analysis to millisecond automated write-back control.
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Appendix D: Full Logic Flowchart and Phase Space Lookup Table

D.1 Full Logic Flowchart

Refer to the master logic schematic tracing the full loop: Observer Boundary Partition $\Pi \rightarrow$ Rigid Manifold Preservation $\Pi_{\perp} \rightarrow$ Residual Extraction $\Delta \rightarrow$ Metacognitive Rewriting $\Phi \rightarrow$ Automated Decision Write-Back.

D.2 Holographic Key Operators Lookup Table

Complete lookup matrix mapping physical mechanisms, mathematical formalisms, and neural/silicon implementations across Volumes I to IV.